

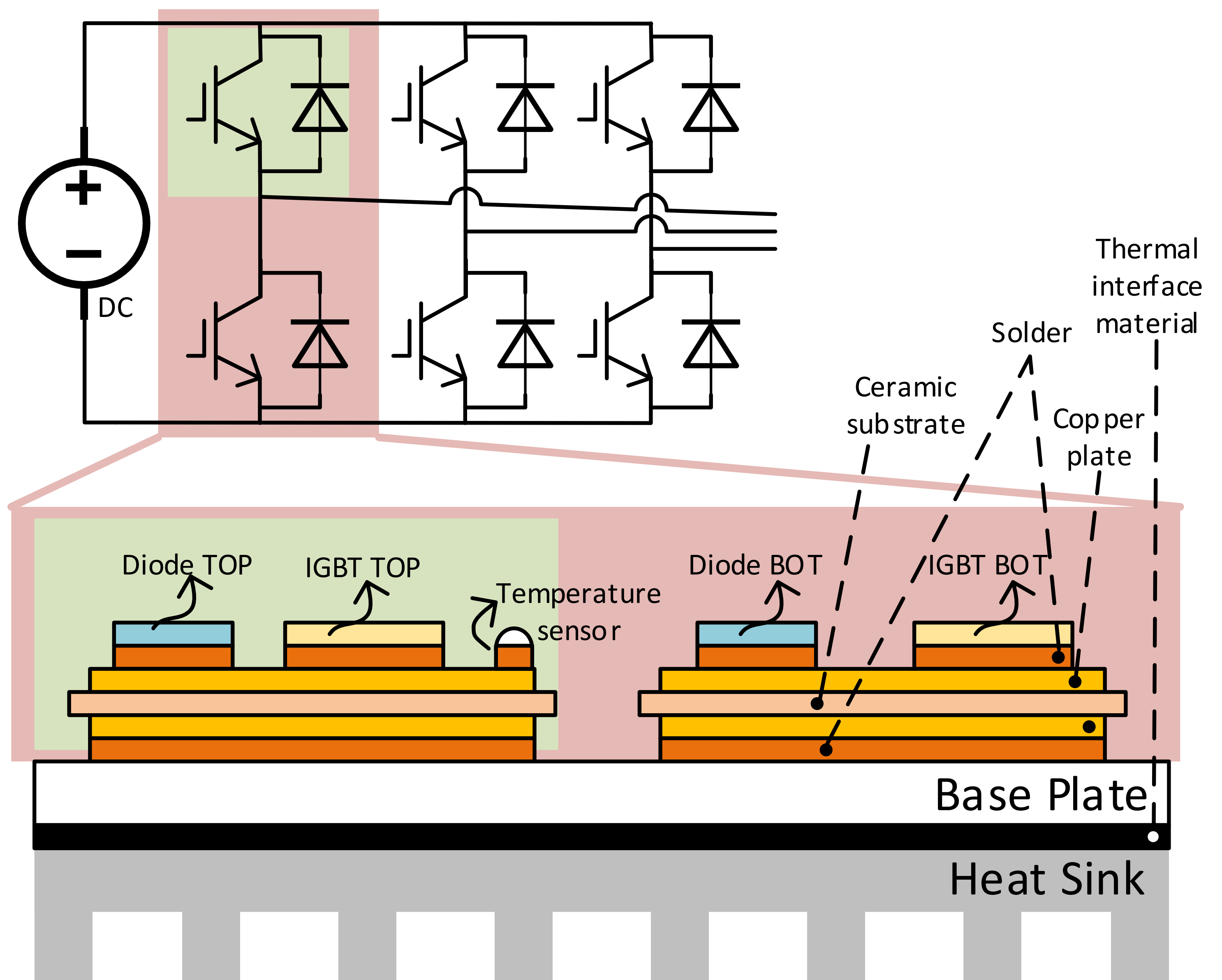
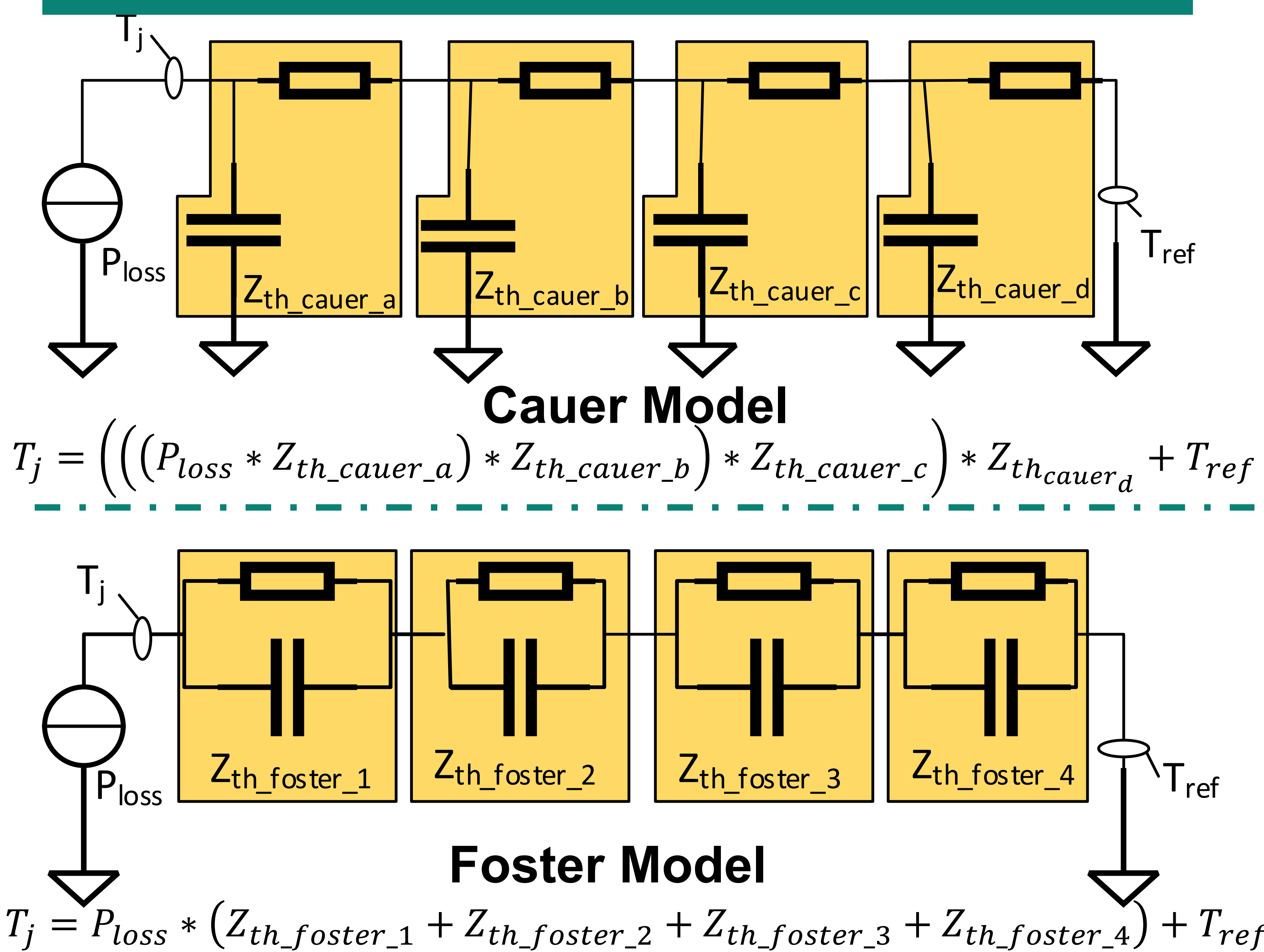
Embedded Implementation of a Thermal Coupling Matrix Using AURIX TC4x Parallel Processing Unit for xEV Applications

Dr. Renke Han, Mohammad Moniruzzaman, Mark Nils Münzer
Infineon Technologies AG, Neubiberg, Germany

Introduction

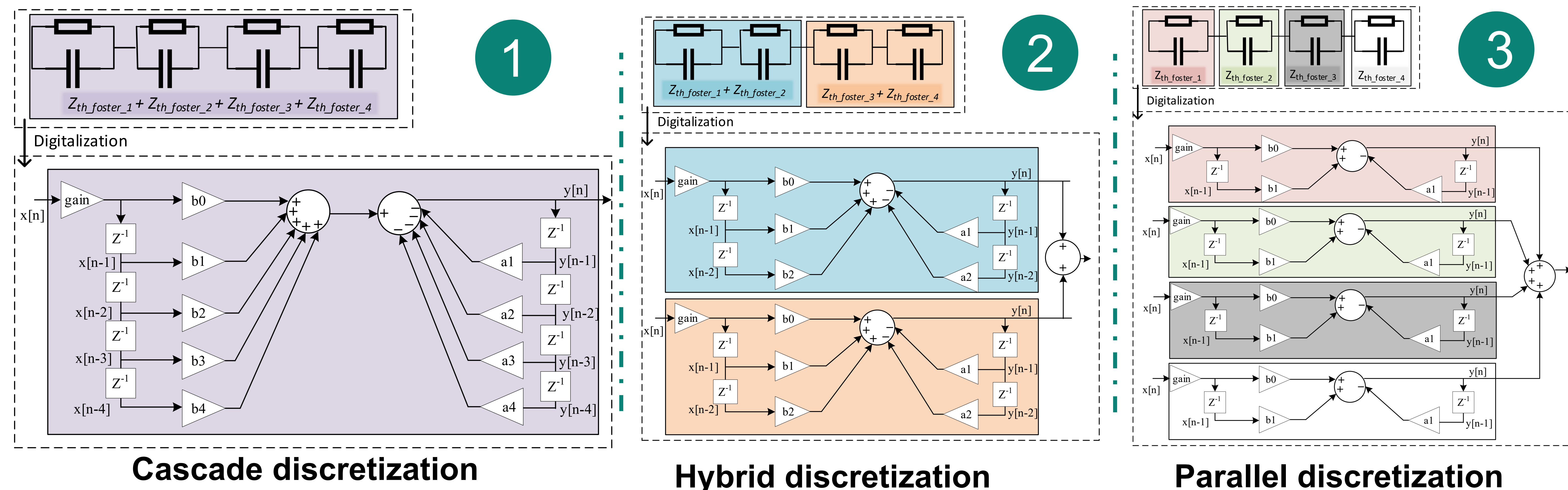
- Real time (us) junction temperature estimation in embedded implementation
- Execute **144 thermal couplings** within **4 us**
- Leverage **AURIX™ TC4x Parallel Processing Unit** with **full flexibility**
- Remove overly conservative system design for xEV applications supporting system cost down

Modeling



Discretization & Digitalization

- Sensitive to parameters;
 - Double floating point might be required;
 - Complex high order digital filter;
- Single precision floating point is sufficient;
 - Simply first order digital filter;
 - Fully automatic implementation workflow possible;

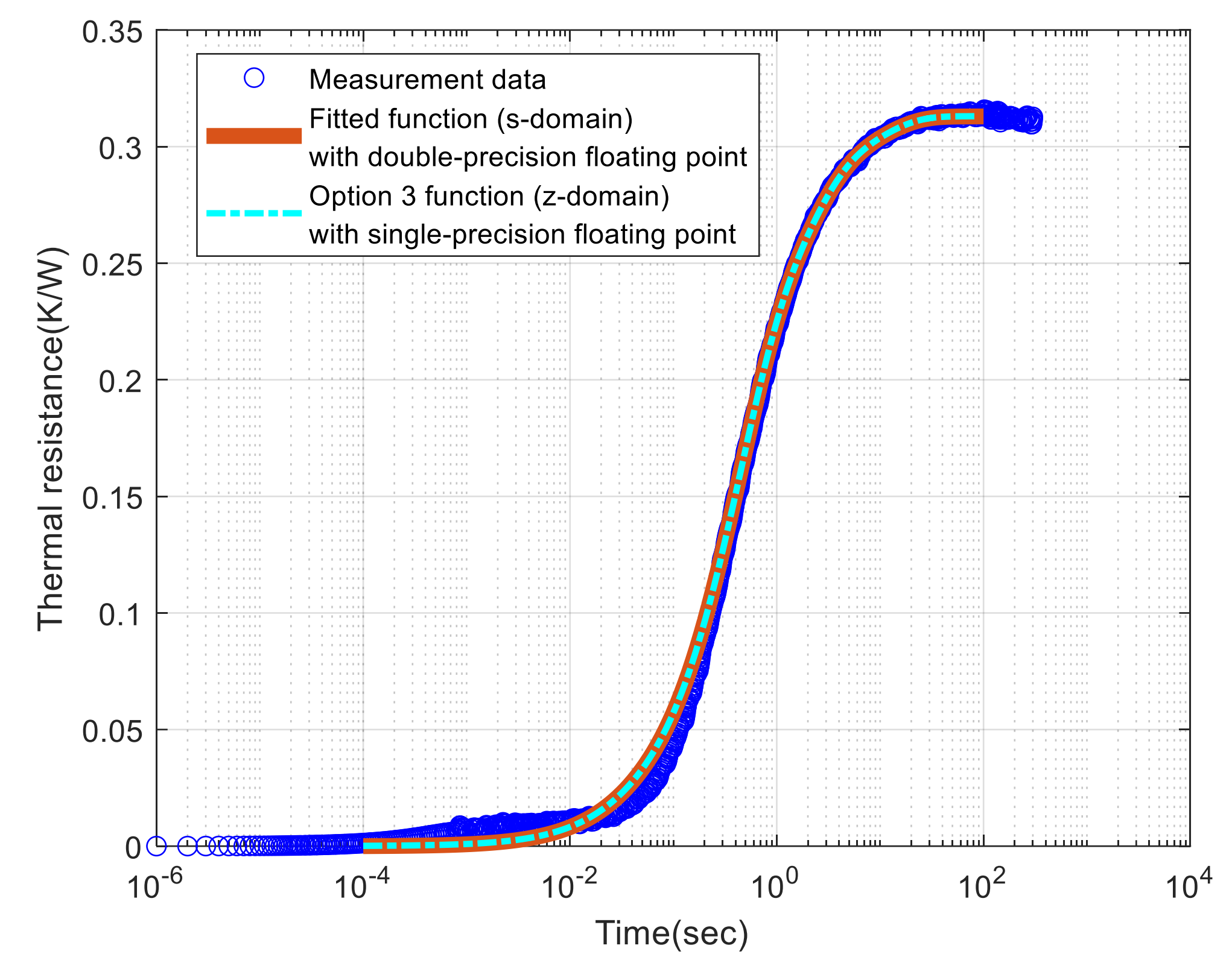
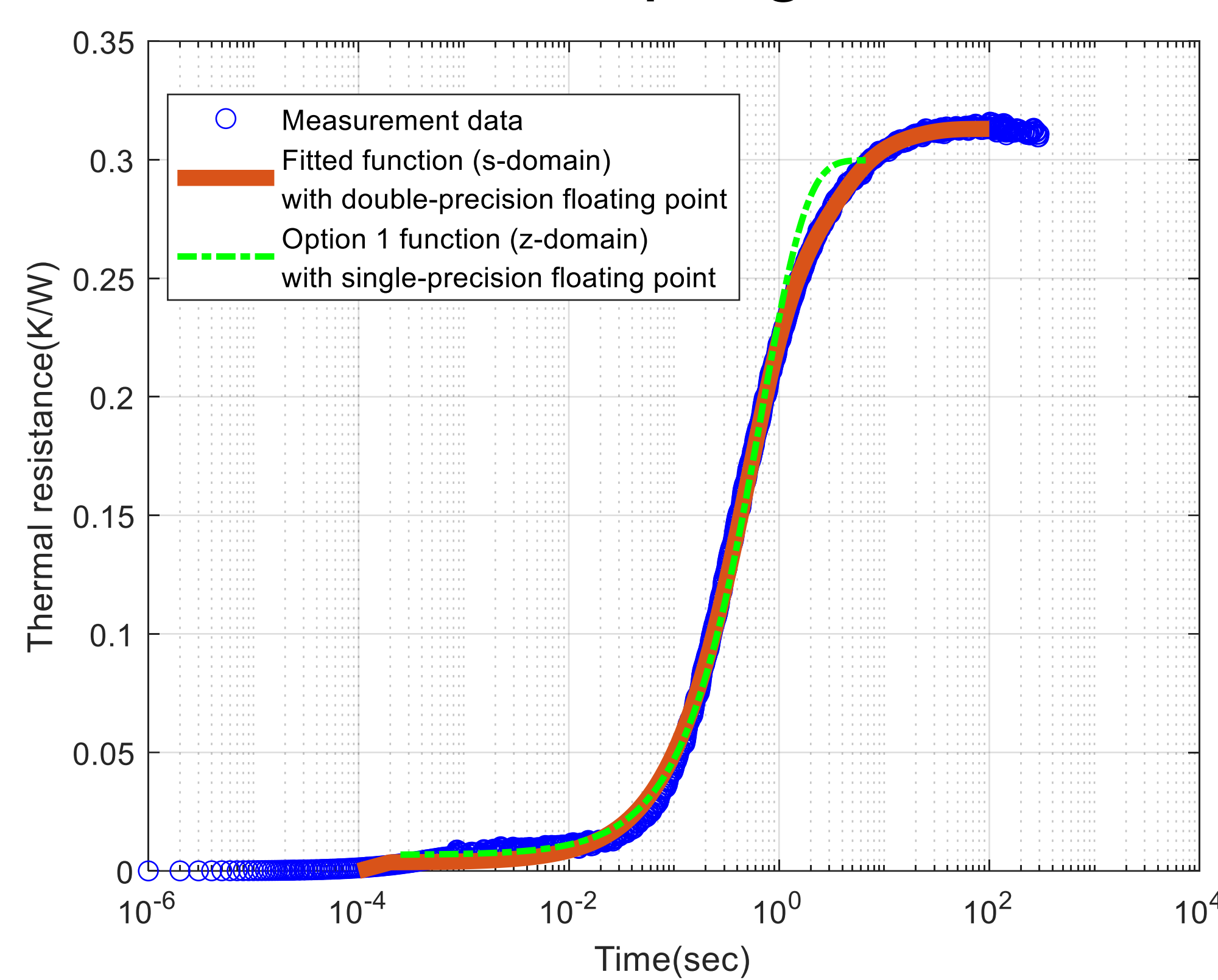
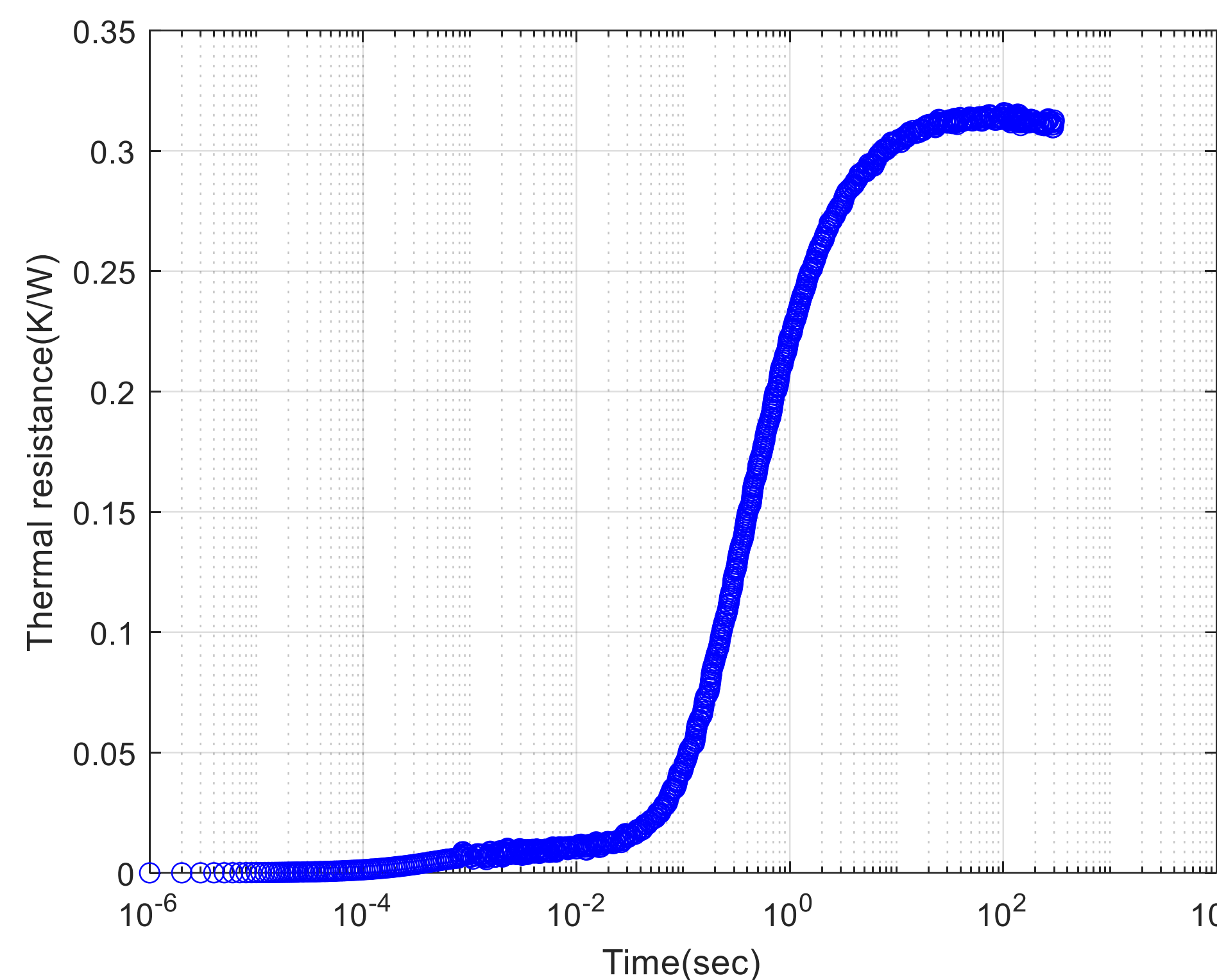


	Set 1		Set 2	
	$R(K/W)$	$\tau(sec)$	$R(K/W)$	$\tau(sec)$
$Z_{th,1}$	0.0298	8.3703	-0.0168	21.5863
$Z_{th,2}$	0.0909	1.6332	-0.0056	2.6107
$Z_{th,3}$	0.1846	0.3685	0.0663	0.2987
$Z_{th,4}$	0.00763	0.0231	-0.0070	0.0091

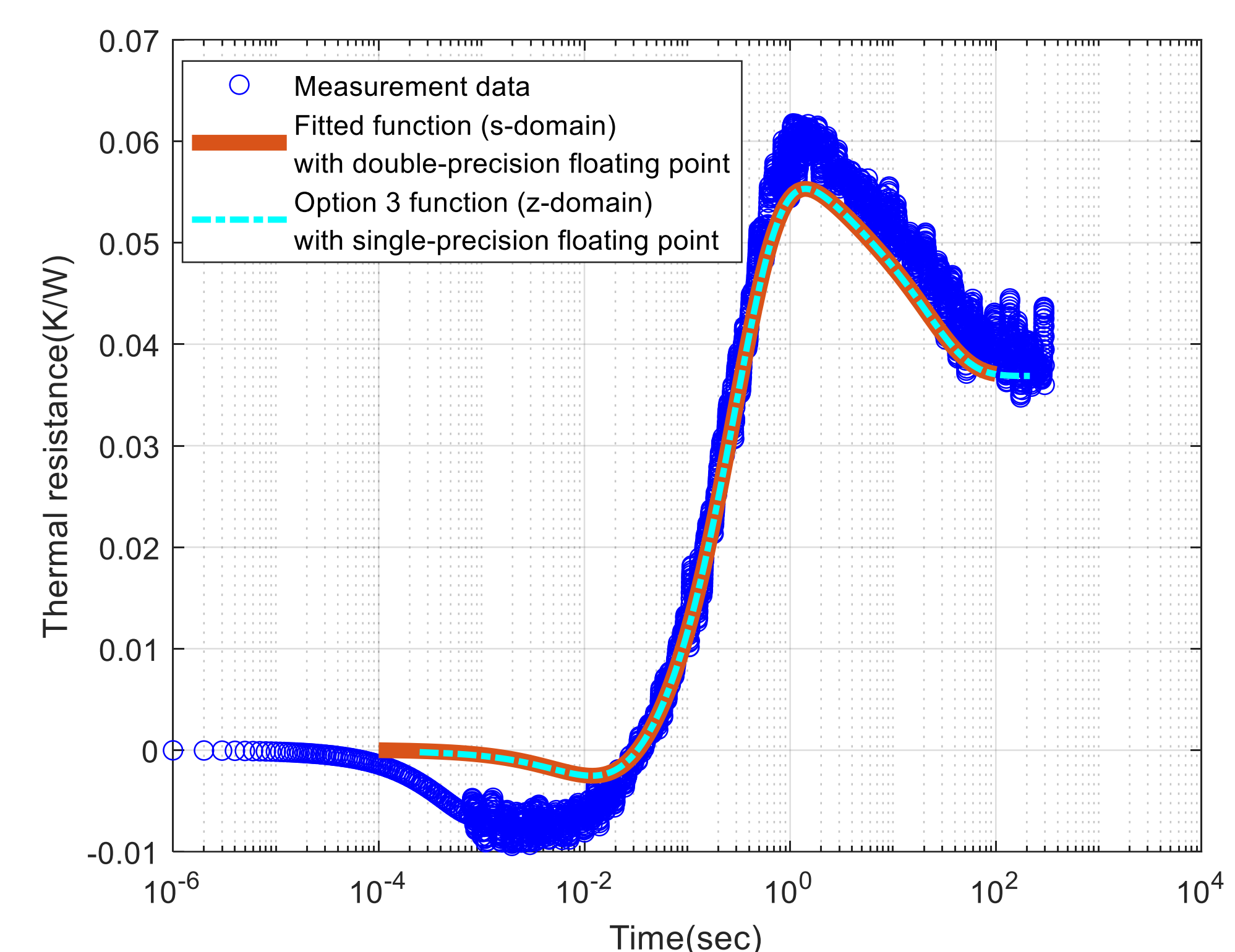
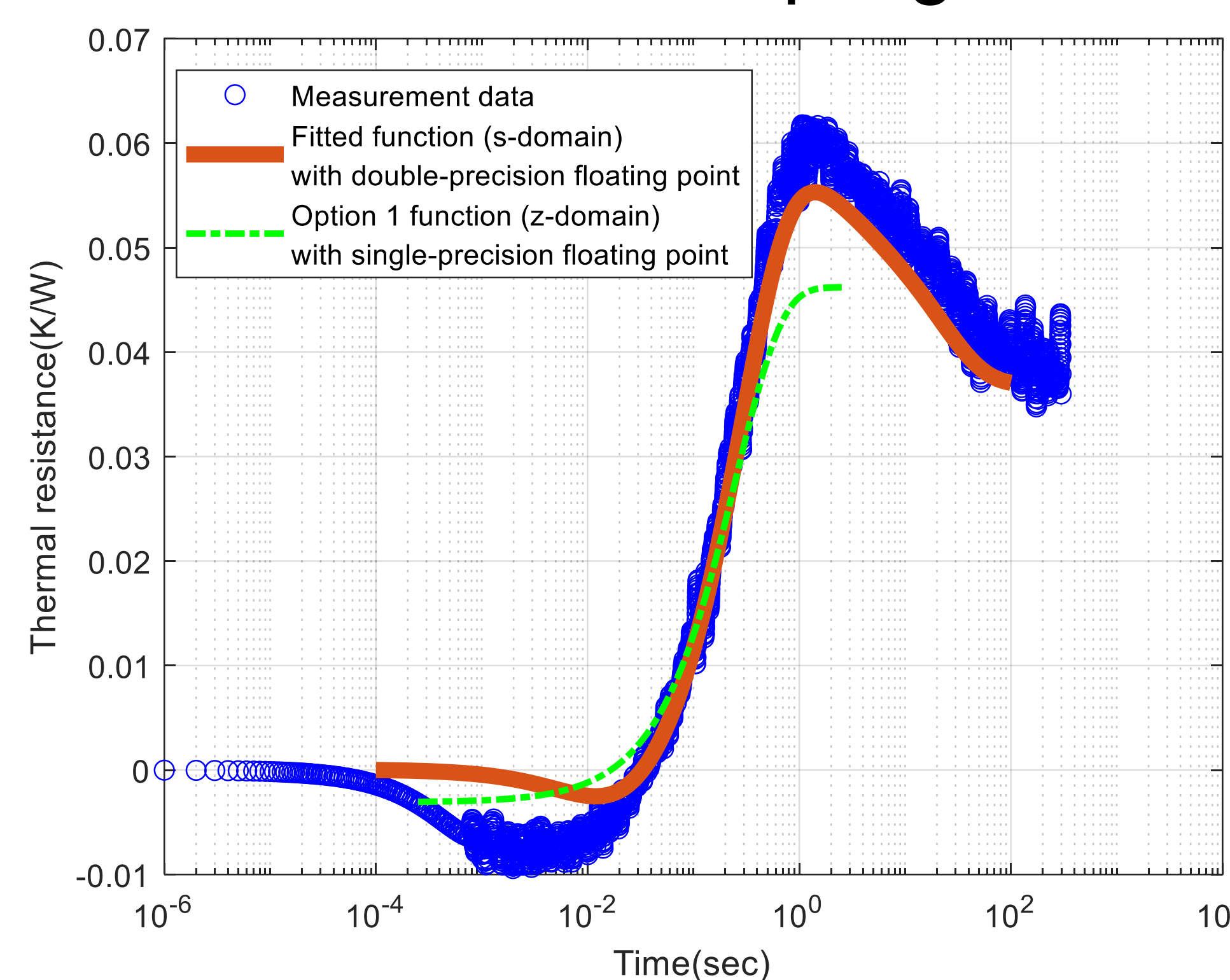
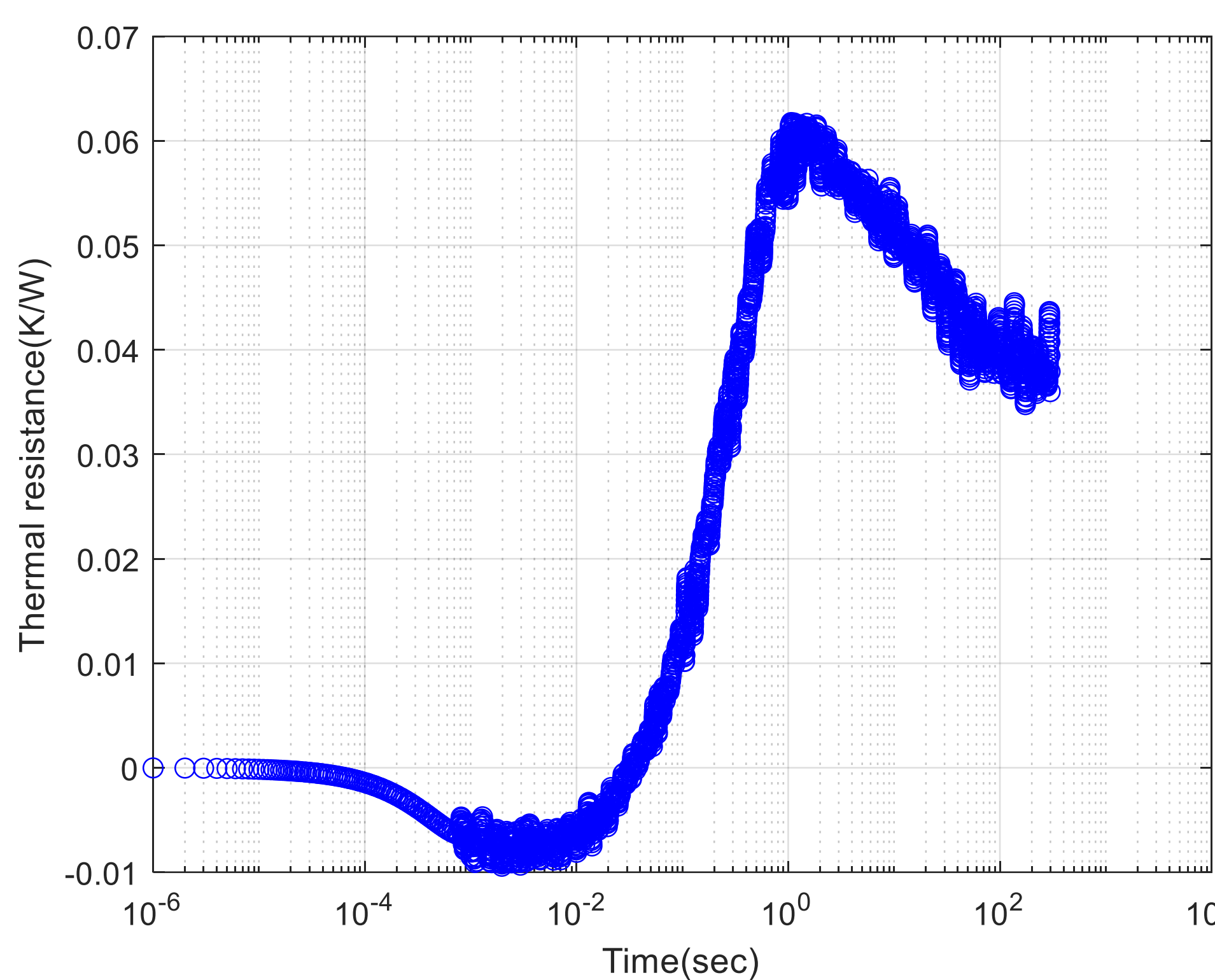
Set 1 is normally self-heating thermal coupling behavior

- Set 2 with under/overshoot is normally cross coupling behavior
- Set 2 time constant among different impedances are more than three orders of magnitude (challenge!)

Thermal coupling set 1

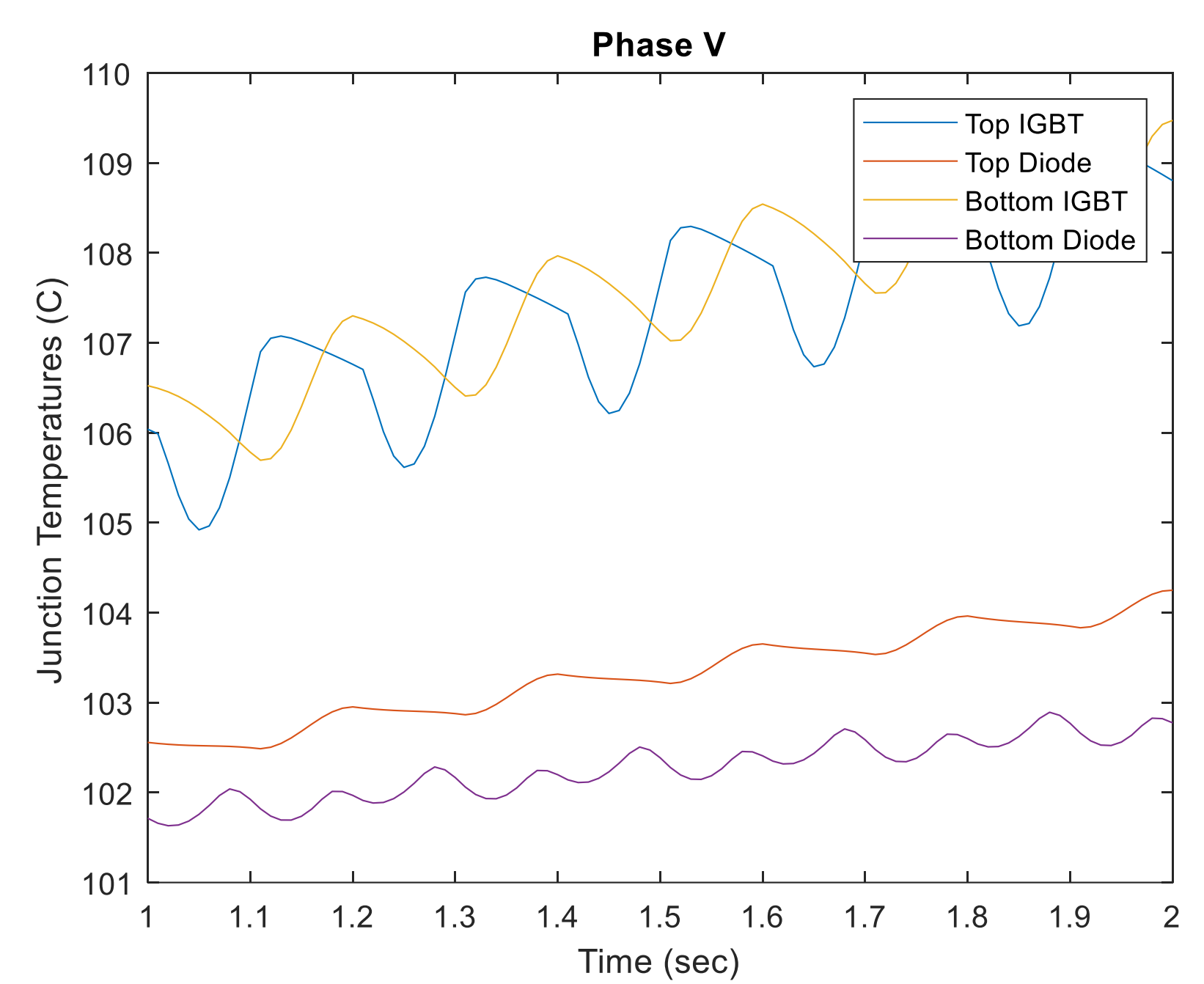
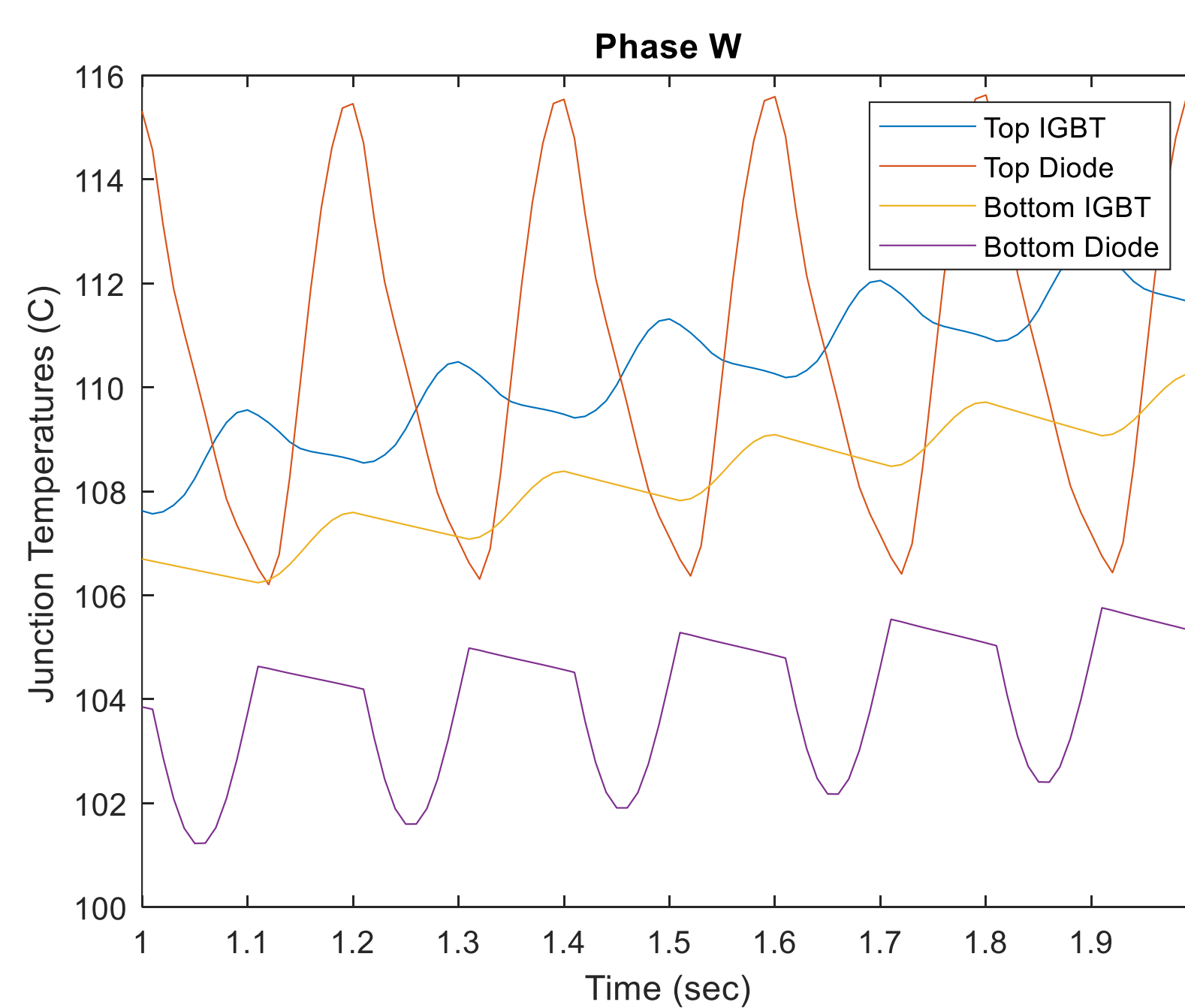
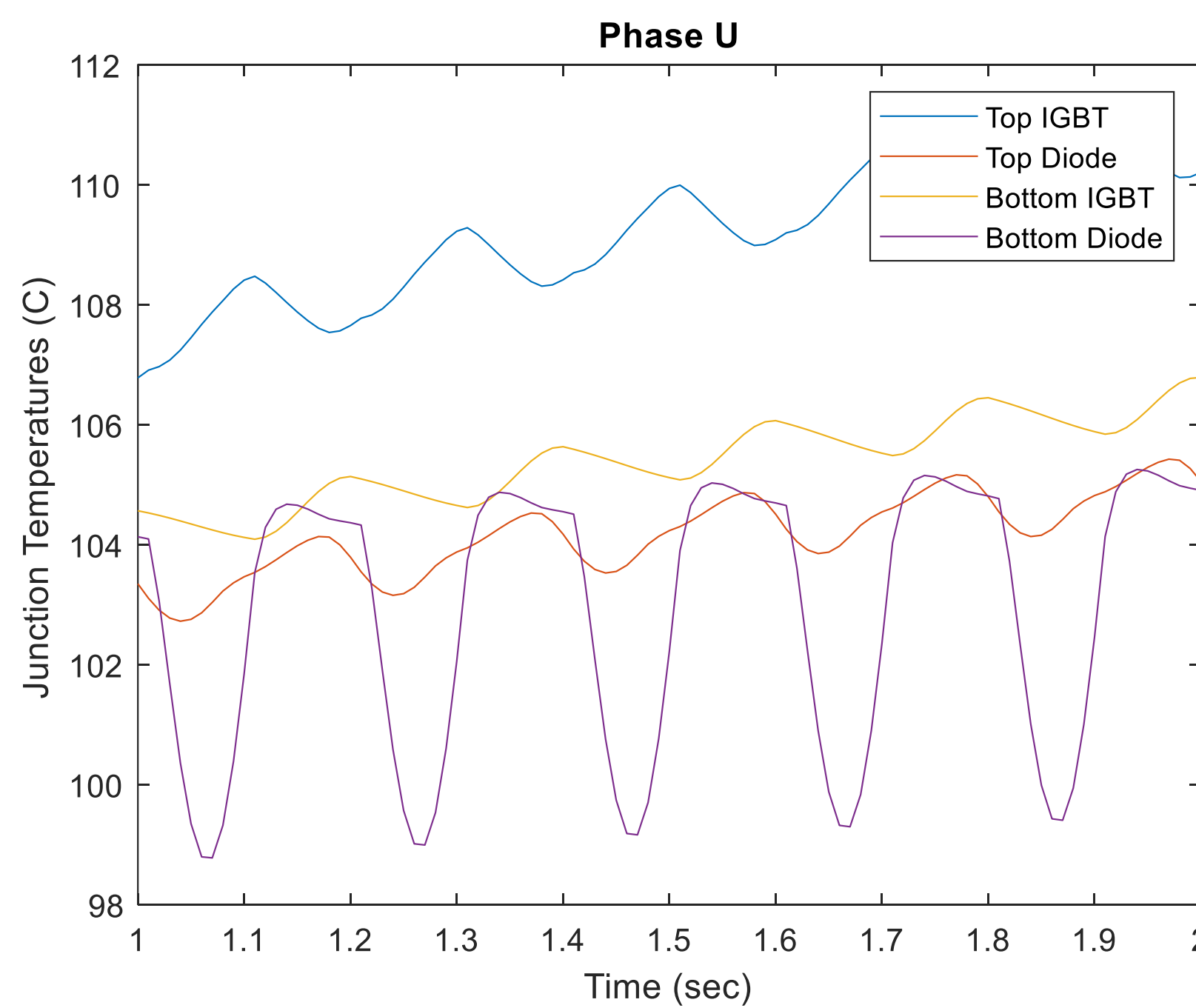
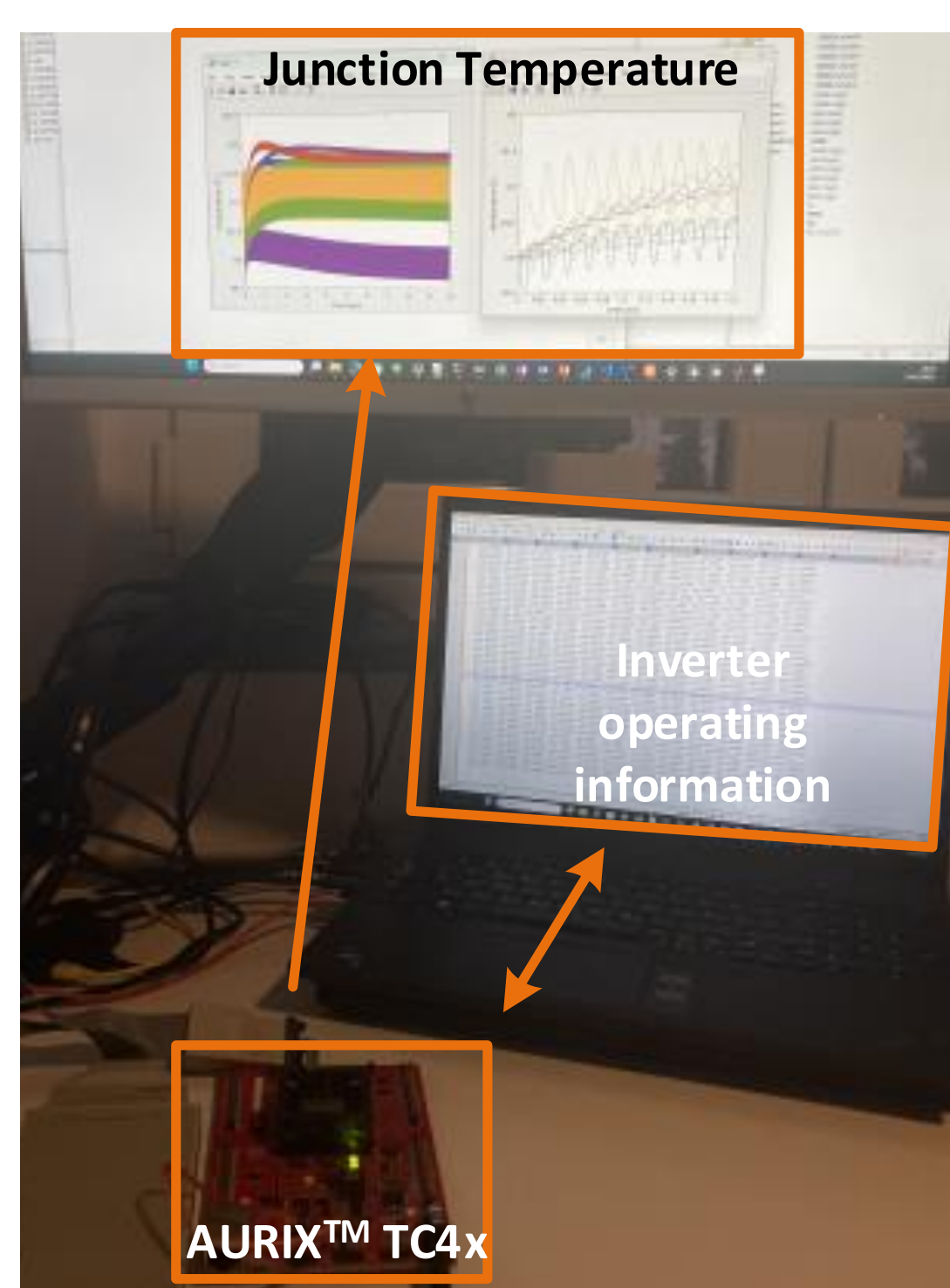


Thermal coupling set 2



Thermal behavior demo

144 thermal coupling with
3.71 us execution time



	Core frequency	Execution Time
TC3x TriCore	300 MHz	35.01 μ s
TC4x TriCore	400 MHz	26.26 μ s
TC4x PPU	400 MHz	3.71 μ s

Practical Consideration

- Accuracy:** the proposed workflow can nearly reproduce the calibrated thermal characteristic with nearly 100% fidelity
- 1ms updated rate is sufficient** because intrinsic properties of semiconductor inherently limit the speed of thermal transients
- System integration:** AURIX PPU can provide sufficient headroom to embed this concept into mass-production solution, e.g., including more temperature points

More use cases about PPU

Online webinar: **Boosting OEM profitability: AURIX™ TC4x PPU transforms your e-drivetrain**

Scan

